

Final Report Topic: Religion, Division of Labor, and Conflict: Anti-Semitism in Germany over 600 Years

In this topic, you will analyze data from an empirical study by Becker & Pascali (2019) (“BP” from now on) on historical causes and mechanisms behind antisemitism in Germany. BP study rich city and county level data spanning over 600 years from German and Prussian administrative and other sources. The paper asks the question of whether Protestantism caused antisemitism in historic German regions.

The theory behind the article is that, whenever ethnic/religious or other groups face increased economic competition, they are more likely to engage in conflict compared to a situation where they specialize and, thus, act more like economic complementarities than substitutes.

They exploit the fact that Catholicism had a strong dogma against usury, including against lending at interest, while Jewish minorities were allowed to lend money and engage in other finance/trade related activities that were considered banned by Catholic dogma. Moreover, Jewish communities often had higher levels of human capital (literacy etc.) compared to their Christian counterparts. Thus, overall, there was a significant specialization of Jews in finance, trade and related sectors of the pre-Reformation German economy. After the Reformation however, cities or regions that converted to Protestantism were no longer subject to the same type of restrictions as Catholics. Moreover, there was also an increase of human capital in Protestants relative to Catholics, partly due to increases in reading investment and education (see Becker and Woessmann 2009). Both led to an increased influx of Protestants into finance/trade and, thus, higher economic competition between Protestants and Jews compared to regions that remained Catholic.

The BP study has two main parts. We focus on a subset of their analysis. In particular, you are supposed to investigate the following research questions:

1. Did Protestantism have long-lasting effects on the prevalence of antisemitic political parties in the German Second Reich?
2. Did Protestantism cause increase in Pogroms relative to Catholic regions & did this difference increase whenever economic competition between Jews and Christians was more pronounced?

You use two different data sets to address this question:

- For question 1, you use STATA format data set `prussia.simple.dta` which contains 452 observations with regional variations in religion, literacy rates, school enrollment, and more. The units of observation are the Prussian Kreise (counties) as of 1871. The variables are contained in Table 1 below.
- For question 2, you use city-level panel data over centuries `pdata.city.R`. This is a `plm()` panel data frame that can be loaded via the `readRDS()` command. All variables are listed in Table 2 below.

General Hints: Read the paper and discuss the broad research design together. In particular, focus your reading on: Section I.1, Sections II.A & II.C, Sections III.A & III.C, and Section IV.A.

For question 1, try to replicate Table 4, Column 3. Provide additional analysis (see keywords). You may/should go quite a bit beyond the content of the paper and provide additional descriptive evidence as well as other research designs and tests that complement your analysis. For that, you may choose to do the analysis based on continuous and/or binary treatment and instrument as available in the data.

For question 2, try to replicate Table 6, Columns 3 and 4. You may deviate from the research design and conduct additional robustness checks and specification tests/analysis.

For all questions, provide definitions of the research questions and target parameter(s) using causal formalization, assumptions + intuition and/or proof of identification. At each step, be precise in your notation and formalization.

Additional Keywords for Research Question 1: Regression. Matching. Weighting. Balance. Instrumental Variables. First stage. LATE. Clustering.

Additional Keywords for Research Question 2: Parallel trends. Pre-trends. Differences-in-Differences. Fixed Effects. Instrumental-Variables Differences-in-Differences. Clustering.

Main reference: Becker, S. O., & Pascali, L. (2019). Religion, division of labor, and conflict: Anti-Semitism in Germany over 600 years. *American Economic Review*, 109(5), 1764-1804.

Optional reference: Becker, S. O., & Woessmann, L. (2009). Was Weber wrong? A human capital theory of Protestant economic history. *The Quarterly Journal of Economics*, 124(2), 531-596.

Becker and Woessmann (2009) analyze the data from 19th-century Prussia to investigate the relationship between Protestantism, education, and economic development. The data is similar to the Prussian data used for this report. Their goal is to analyze the mechanism behind observed economic prosperity in Protestant regions. In particular, they evaluate the direct effect of Protestantism on economic prosperity and indirect effects through human capital accumulation (education, literacy etc). **It contains a more detailed discussion of the use of *distance to Wittenberg* as an instrument for Protestantism** which might help your discussion.

Table 1: Variable Definitions for County-Level Data

Variable Name	Description
share_antisem_reich_1898	Share of votes for anti-Semitic parties (1898)
f_prot_1882	Share Protestants 1882
kmwittenberg	Distance to Wittenberg in km
f_young	Share age below 10
f_fem	Share females
f_ortsgeb	Share born in municipality
f_pruss	Share of Prussian origin
hhsz	Average household size
lnpop	ln(Population size)
posen	If county in Posen
f_urban	Share of county pop. in urban areas
code_reichstag_wk	Precinct ID
prot_majority	= 1 if <code>f_prot_1882</code> > 0.5 and 0 else
wittenberg_close	= 1 if <code>kmwittenberg</code> < 500 and 0 else

Table 2: Variable Definitions for City-Level Panel

Variable	Description
id	City identifier used in the panel (unit is city $\times$ century).
pogrom_small_t	Indicator = 1 if the city-century shows any expulsion or killing of Jews.
century_14001500	Dummy for the 1400–1500 century.
century_15001600	Dummy for the 1500–1600 century.
century_16001700	Dummy for the 1600–1700 century.
century_17001800	Dummy for the 1700–1800 century.
post_protestant_1546	Protestant city (ruler Protestant in 1546) $\times$ post-Reformation dummy.
PostXJLend_any_geq2	Post-Reformation $\times$ indicator for pre-1500 Jewish lending.
PostXJLend_anyXProt	Triple interaction: Protestant $\times$ post-Reformation $\times$ pre-1500 Jewish lending.
post_n_trade1serv_1500	Post-Reformation $\times$ number of pre-1500 trade & service sectors.
pr_post_n_trade1serv_1500	Protestant $\times$ post-Reformation $\times$ number of pre-1500 trade & service sectors.
Optional Variables	Description
school_begincent	Indicator = 1 if a primary or secondary school is recorded in the city during century.
involvement_elsewhere	Indicator = 1 if the city was involved in a war elsewhere during the century.
occupied	Indicator = 1 if the city was occupied during the century.
Jewish_presence_imp	Indicator = 1 if there is evidence of a Jewish community in the city during century.
PXnearbattle	Interaction: post-Reformation dummy $\times$ indicator for a nearby battle.
PXinvolvement_elsewhere	Interaction: post-Reformation dummy $\times$ war involvement elsewhere.
post_p_all_1500	Post-Reformation $\times$ pre-1500 lending sectors.
time_all	Combined numeric time variable covering all centuries (only used for setup of panel).